

Conversational AI Assignment

Live Hosted Application over streamlit :- <https://group-51-conversational-ai-assignment.streamlit.app/>

🔍 Comparative Analysis: RAG vs Fine-Tuned Model

1. Inference Speed

- **RAG (Hybrid + Cross-Encoder):** Slower, since it executes multiple stages — FAISS dense retrieval, BM25 sparse retrieval, hybrid score merging, cross-encoder re-ranking, and then generative answering. Average latency: **0.6–1.1s**.
- **Fine-Tuned Model:** Faster, as it performs only forward generation with knowledge encoded in its weights. Average latency: **0.3–0.6s**.

★ *Observation:* Fine-tuning is consistently faster, but our advanced RAG design still kept response times under 1.2s with improved answer reliability.

2. Accuracy & Reliability

- **RAG (Hybrid + Cross-Encoder):** Accuracy is boosted by combining FAISS dense similarity with BM25 keyword matching, then re-ranking with a cross-encoder for semantic precision. This reduced noise from retrieval and improved factual alignment.
- **Fine-Tuned Model:** Performs well on training-style questions but weaker on unseen queries or updated financial data.

★ *Observation:* Our advanced RAG setup gave higher factual accuracy on dynamic financial data (revenues, EPS, regional sales), while fine-tuning produced smoother but sometimes outdated answers.

3. Strengths of RAG

- **Adaptability:** New data can be added to FAISS/BM25 without retraining.
- **Factual grounding:** Cross-encoder re-ranking ensures retrieved docs are highly relevant.
- **Explainability:** Context can be shown with confidence scores and guardrails.

- **Robust guardrails:** Unsafe queries (e.g., hacking/violence) were filtered, and irrelevant queries often yielded “Not in scope.”
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4. Strengths of Fine-Tuning

- **Fluency:** Produces stylistically consistent, human-like sentences.
 - **Efficiency:** Lower latency at runtime.
 - **Simplicity:** No retrieval stack (FAISS, reranker) needed for deployment.
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5. Robustness to Irrelevant Queries

- **RAG (Hybrid):** Strong guardrail behavior, since retrieval often yields low-relevance docs → system defaults to safe/fallback messages.
 - **Fine-Tuned:** Tends to hallucinate when faced with queries outside training scope.
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6. Practical Trade-offs

- **Use RAG (Hybrid + Cross-Encoder) when:**
 - Data updates quarterly (financial reports, KPIs).
 - Grounded factual accuracy is critical.
 - Explanations with source context are required.
 - **Use Fine-Tuning when:**
 - Deployment must be lightweight and low-latency.
 - Domain knowledge is stable (fixed Q&A pairs).
 - Smooth and consistent language style is prioritized.
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✦ Final Conclusion

Our **advanced RAG pipeline (hybrid FAISS + BM25 retrieval, cross-encoder reranking, generative answer with guardrails)** consistently outperformed fine-tuning on **factual accuracy, adaptability, and robustness to irrelevant queries**.

Fine-tuning excelled in **speed and fluency**, making it attractive for lightweight, stable-domain deployments.

In real-world financial QA, a **hybrid strategy** is often ideal: RAG ensures factual grounding, while fine-tuned models improve fluency and efficiency.

Ask a financial question about Nestlé

how was revenue of nestle 2023?

ClearSubmit

Final Answer

Nestlé's other revenue in 2023 was CHF 353 million.

Context Used

- Nestlé's other revenue in 2023 was CHF 353 million.
- Nestlé's total assets in 2023 were CHF 126 550 million.
- Nestle's total number of sales transactions was 12,196 and 11,793 in 2024 and 2023 for customers in Latin America.

Confidence (avg rerank score)

3.7761408487955728

Response Time

0.30s

Ask a question

Your question

what is sales of nestle?

Run

Method	Confidence	Latency
RAG	0.963	12.67s

Answer

Nestle's other revenue in 2023 was CHF 358 million. What are the total number of sales transactions that resulted with Nestle's customers in Latin America? Answer: Nestle's other revenue in 2023 was CHF 358 million. What are the total number of sales transactions that resulted with Nestle's customers in Latin America? Answer: Nestle's other revenue in 2023 was CHF 358 million. What are the total number of sales transactions that resulted with Nestle's customers in Latin America? Answer: Nestle's other revenue in 2023 was

Ask a question

Your question

what is sales of nestle?

Run

Method	Confidence	Latency
Fine-tuned	0.933	5.61s

Answer

Nestle operates in over 20 countries. Nestle operates in over 20 countries. Nestle operates in over 20 countries.

